WTI forecasting with an LSTM

QF5210 | Individual model contribution

Executed code, out-of-sample evidence, technical methodology and presentation guide

Data: 4 August 2011 - 6 October 2026
Test origins: 2 January 2024 onward
Target: nominal WTI Cushing spot price, USD/barrel

Main finding

At horizon 60, LSTM reduces RMSE by 8.1% against published-price persistence. MAE barely changes; ridge performs better, and gains are not consistent across years. Approximate bootstrap ranges include zero.

Use the accompanying editable slides for the short group talk. This report supplies the details needed to explain, reproduce and assess the LSTM contribution.

Results from the supplied data

Main conclusion

The multivariate LSTM provides limited and regime-dependent forecasting gains. It reduces horizon-60 RMSE by 8.14% relative to published-price persistence, but its horizon-60 MAE is almost unchanged. Ridge regression is stronger on most of the key comparisons. The evidence does not support saying that LSTM is the best group model or that it produces a reliable trading advantage.

The selected model uses a 60-observation lookback, 32 hidden units, dropout 0.2, AdamW learning rate 0.001, weight decay 0.001, batch size 128, and four final training epochs. The final forecast is the equal-weight average of seeds 42, 43 and 44. The four epochs were selected by validation early stopping, not by test performance.

Forecast accuracy

All MAE/RMSE values are USD per barrel. Positive improvement means smaller error than persistence. Each horizon uses every test origin with an observed target for that horizon; within a horizon every model uses identical rows.

Model	Horizon	N	MAE	RMSE	RMSE gain (%)
LSTM	1	689	3.726	5.485	-0.246
Persistence	1	689	3.697	5.471	0.000
Ridge	1	689	3.697	5.527	-1.023
LSTM	5	685	4.360	6.639	0.621
Persistence	5	685	4.363	6.681	0.000
Ridge	5	685	4.343	6.652	0.440
LSTM	20	670	6.784	9.509	2.839
Persistence	20	670	6.558	9.787	0.000
Ridge	20	670	6.278	9.287	5.105
LSTM	60	630	10.336	14.294	8.140
Persistence	60	630	10.346	15.561	0.000
Ridge	60	630	9.240	13.955	10.316

Horizon-specific counts differ because later targets are unavailable near the end of the file. `test_metrics_common_origins.csv` repeats the comparison using the same 630 origins across all 60 horizons. Use that file when the question is how accuracy changes with horizon under a fixed origin sample.

Directional performance

Horizon	Direction (%)	Up precision	Up recall	Up F1
1	48.110	0.492	0.709	0.581
5	51.241	0.517	0.773	0.620
20	51.045	0.500	0.881	0.638
60	49.285	0.466	0.866	0.606

Direction means whether the future target is above or below the last published WTI quote known at the forecast origin. It is not conventional next-close return direction because yesterday's target is not assumed available in this dataset. Actual zero changes are excluded from directional accuracy, and their counts are reported. Persistence predicts no movement and abstains from a directional view, so its 0% score under the strict sign rule is not evidence that LSTM beats an always-up/down classifier. High up-recall alongside modest precision can reflect an upward bias.

Uncertainty around improvement

Horizon	Lower (%)	Upper (%)	Block length	Resamples	N
1	-2.488	0.856	60	1000	689
5	-2.421	2.277	60	1000	685
20	-13.039	7.119	60	1000	670
60	-11.723	13.894	60	1000	630

These are approximate 95% paired moving-block bootstrap ranges for percentage RMSE improvement, using 1,000 circular resamples and 60-origin blocks. Every range includes zero. For horizon 60, the range is approximately -11.72% to +13.89%. Report this as uncertain improvement, not statistical proof. Overlap, nonstationarity, block choice and a small effective number of blocks limit inference; this is not a formal multiple-testing-adjusted test.

Robustness and ablation

Grouping	Group	N	RMSE	RMSE gain (%)
origin_year	2024	250	7.165	-10.148
origin_year	2025	248	11.871	-3.305
origin_year	2026	132	24.763	13.915
origin_volatility	high_OVX	327	15.574	12.854
origin_volatility	low_OVX	303	12.769	-1.340

The horizon-60 improvement occurs mainly in 2026 and high-OVX origins. The LSTM is worse than persistence for origins in 2024 and 2025. Groups are defined by the forecast origin, so a 2025 origin can have a 2026 target. OVX is an observable uncertainty proxy, not an ex-post realized-volatility classification; the pre-test median threshold is 34.855.

The published-WTI-only LSTM is a useful ablation. At horizon 60, its RMSE is 14.863 and MAE is 9.756, compared with the multivariate ensemble's 14.294 and 10.336. Additional predictors therefore do not improve every metric. The ablation uses one seed and the main model uses three; this is a descriptive comparison, not an isolated causal estimate of the value of each predictor. No feature causality or importance claim follows from it.

Last supplied forecast origin

Origin date	Horizon	Anchor	Forecast	Seed min	Seed max
2026-10-06	1	85.230	84.373	83.518	85.575
2026-10-06	5	85.230	83.726	82.944	84.188
2026-10-06	20	85.230	82.524	79.208	84.835
2026-10-06	60	85.230	78.363	73.635	83.821

This is a historical forecast issued conceptually before 6 October 2026, using that row's pre-midnight information. Horizon 1 targets 6 October itself; horizon 2 targets the next observed trading date. The actual horizon-1 value of 96.24 was not used as input, and the forecast of 84.37 illustrates a substantial miss. Do not present this as a current forecast or use the actual target to repair it. Future target dates are deliberately blank because no future observation calendar was supplied. Seed minima/maxima quantify sensitivity to initialization; they are not prediction intervals.

Technical methodology and code guide

1. Scope and relationship to the project brief

The relevant assignment is the oil forecasting alternative on PDF pages 8-9 (printed slide numbers 32-33). It asks for daily crude oil price and/or return forecasts for 1-60 days, out-of-sample comparisons, multiple accuracy measures, robustness, data documentation and automatic forecast updates. Economic-value analysis is encouraged, rather than required. The presentation instructions on PDF page 10 specify eight minutes plus two minutes of Q&A per group, participation by every member, and a zipped submission containing editable slides, data and code by 31 October 2026. Check Canvas for any changes or additional instructions.

This contribution covers the deep-learning component independently. The question is: does a sequence model of already published energy and financial variables improve WTI forecasts over an available-information persistence benchmark? Other teammates can compare their ARIMA/VAR/tree models on the exported date pairs, or use these LSTM predictions in an ensemble with weights chosen on validation data. No teammate outputs feed this LSTM.

2. Data audit and provenance

The source is the supplied `wti_daily.csv` plus its supplied `features.csv` dictionary. The data has 3,790 rows, from 2011-08-04 through 2026-10-06, 32 predictors, one date and one target. There are no missing cells or duplicate dates. Dates are already increasing and remain in that order. All predictors must be finite; the script raises an error rather than silently backfilling or deleting rows. The original input CSVs are copied unchanged under `data/`.

The target is nominal WTI Cushing spot price, USD/barrel, identified in the dictionary as FRED DCOILWTICO (underlying EIA source). Predictors include Brent and refined-product prices, inventories, imports/exports, production and demand proxies, plus rates, equity prices, implied volatility and the dollar index. The full variable-level units, source IDs and release assumptions remain in `data/features.csv`; that file is part of the submission.

The input SHA-256 is `8431640f4cf19f67aed661dce306103cc69f5d08d7a32143ad5f43152e4913eb`. This identifies the exact CSV used, rather than proving correctness. The GitHub repository URL, commit, raw downloads, release timestamps and preprocessing code were not supplied. Availability claims are inherited from the teammate's dictionary and cannot be independently certified from these two files. FRED states data are subject to revision. This is a chronological backtest conditional on the processed alignment, not a fully audited real-time vintage study.

One observation is nonpositive; the minimum is -36.98. Keep it. We do not log-transform prices, calculate log returns across zero, or replace the observation. Prices and changes remain meaningful in dollars even when negative. MAPE is supplementary and explicitly restricted to actual prices greater than USD 1; its eligible sample count is exported. MAE/RMSE over all eligible observations are the headline metrics. The current test sample is positive, but future data and training need not be.

3. Information set and the forecast clock

The dictionary defines row t as the trading day being forecast, with every feature public before 00:00 New York time that day. Weekly EIA observations have already been aligned using a report-release-plus-one-federal-business-day rule. Financial variables use two federal business days under the teammate's assumptions. Repeating a weekly value across daily rows represents the information last available, not fresh daily measurements.

The implementation trusts this alignment and does not shift the predictors again. It excludes `target_wti` completely from input features, including lagged versions. In particular, it does not assume y[t-1] is published just because its economic date precedes t. The anchor a[t] is the predictor `wti_spot_last_published`, which is already available on row t. It can be stale.

For horizon h in 1,...,60, the target index is t+h-1. A 60-row input window runs from t-59 through t. Horizon 1 is the upcoming price on the origin date, viewed before that trading day begins; horizon 60 is 59 subsequent observations later. The horizons count observed trading rows, not calendar days. Missing holidays are not invented, and Friday's h=2 means the next observed date. This convention needs explicit agreement across the group. An after-close origin convention would require shifting both origins and labels consistently.

For example, the last supplied row is 2026-10-06 with an available published WTI anchor of 85.23 and an unavailable-at-origin target of 96.24. The model may read 85.23 and the other predictors, but not 96.24. Its horizon-1 forecast is 84.37, and that error remains in the evaluation.

4. Chronological split and label purging

There is no random train/test split. Nominal training ends 31 December 2021; validation origins start 3 January 2022; test origins start 2 January 2024. A conservative ten-observation release buffer is withheld before both split boundaries. This is a stated implementation assumption, not a verified release schedule for every historical observation.

During model selection, the last eligible training target date is 2021-12-16. Since each sample has 60 labels, the last training origin is 2021-09-22. There are 2,474 common training origins across all candidates. Validation origins run from 2022-01-03 through 2023-09-19, 430 full paths, with last labels on 2023-12-14. A validation origin near December 2023 cannot be used if its 60-day label path crosses into 2024. Purging prevents those test labels entering validation selection.

After configuration and training duration are fixed, the model is reinitialized and refit on all 2,973 pre-test complete-path origins. Final refit labels end 2023-12-14; final weights never consume a 2024-2026 target. This is a frozen-model rolling-origin backtest: each test date supplies a new observed feature window, while model weights/scalers remain fixed. It is not an expanding-window retraining experiment. Later test features are permitted because they would be observed when that later forecast is issued.

Supervised origins stop 59 rows before the label cutoff. Feature scalers are fitted only through the last eligible fit origin; label scalers use only eligible fit samples. This is conservative: some late pre-test features are unused in scaler estimation. Validation/test distributions never determine scaling.

5. Forecast target and preprocessing equations

Let X[t] be the 32-vector of available predictors, y[t] the WTI target, and a[t] the last published quote. First standardize each predictor using pre-test fitted means and standard deviations: z[t,j] = (X[t,j] - mean[j]) / sd[j]. A zero standard deviation is replaced by 1 to avoid division by zero. No whole-sample normalization, interpolation or backward filling is used.

The supervised target is the 60-vector d[t,h] = y[t+h-1] - a[t]. Each horizon has its own training mean m[h] and scale s[h]. The network learns q[t,h] = (d[t,h] - m[h]) / s[h]. The output price is reconstructed as yhat[t,h] = a[t] + m[h] + s[h] * qhat[t,h]. This residual formulation lets persistence correspond to a zero dollar delta while allowing negative prices. It is a dollar change from a stale available quote, not an ordinary close-to-close return.

The objective is the mean squared standardized residual error, averaged equally over samples and all 60 horizons. Horizon scaling prevents high-variance long horizons from dominating solely because their raw dollar errors are larger. The objective is not a trading-loss function, and optimizing it need not minimize every

raw-dollar MAE or directional error.

6. LSTM architecture and intuition

The selected input tensor is [batch, 60 time steps, 32 features]. A one-layer, forward-only LSTM with 32 hidden units processes the sequence. Only the last hidden state enters a dropout layer (0.2) and a linear layer with 60 outputs. There are 10,428 trainable parameters. Hidden and cell states start at zero for every window; they are not carried across batches.

At each step, the forget gate f decides which old memory to retain, input gate i decides how much new information to write, and output gate o decides what memory to reveal. With z as the current standardized feature vector, r as the hidden state, c as cell memory and g as candidate memory:

f = sigmoid(Wf[z,r_prev] + bf); i = sigmoid(Wi[z,r_prev] + bi)

g = tanh(Wg[z,r_prev] + bg); o = sigmoid(Wo[z,r_prev] + bo)

c = f * c_prev + i * g; r = o * tanh(c)

Products are elementwise. These gates can represent nonlinear temporal relationships and persistent information without requiring a fixed AR coefficient per lag. That capability is a modeling motivation, not proof of predictive value. Weekly inputs repeated over a window may add less new information than 60 daily rows suggest.

All 60 horizons are predicted directly from the same known history. The network does not feed a predicted price into tomorrow's inputs or require future macro/inventory values. Direct outputs avoid recursive error propagation but do not impose a smooth or internally coherent future path; some horizon-to-horizon jaggedness is expected.

Dropout is applied after the LSTM. Setting the built-in PyTorch recurrent dropout on a single-layer LSTM would not regularize between recurrent layers; the explicit dropout in `head` avoids that issue. Evaluation uses `model.eval()` to disable dropout.

7. Model selection and final fitting

Three small candidates were predefined: lookback/hidden/dropout of (20,16,0.1), (60,16,0.1), and (60,32,0.2). All use learning rate 0.001, one layer, AdamW weight decay 0.001, batch size 128 and gradient clipping at norm 1. A maximum of 80 epochs and patience 12 control training. A validation improvement must exceed 0.00001 in standardized MSE. Common training and validation origins permit a fair comparison despite different lookbacks.

Candidate	Lookback	Hidden	Dropout	Learning rate	Best epoch	Val. MSE
0	20	16	0.100	0.001	8	1.658
1	60	16	0.100	0.001	7	1.684
2	60	32	0.200	0.001	4	1.560

Candidate 2 was selected at epoch 4. The best validation checkpoint determines the duration; final fitting starts from new initializations and trains for exactly four epochs on combined pre-test data. Three seeds were fixed in advance: 42, 43 and 44. Their price forecasts are averaged equally. Seed-specific errors are retained in `outputs/seed_stability.csv`; the seed with best test performance is never selected. Mini-batch shuffling occurs only among already constructed training windows, after the chronological split. It does not randomize train/test allocation, and independent zero-initialized windows permit it.

The search is deliberately modest because only about 3,000 eligible supervised origins exist and neighboring windows/targets overlap heavily. This is not an exhaustive architecture search. An early selected epoch can

indicate rapid overfitting or distribution mismatch; it does not prove that four epochs are universally optimal.

8. Benchmarks and ablation

Persistence forecasts a[t] at every horizon. It is a no-change forecast based on the available published quote. It is not the stronger conventional y[t-1] random walk unless yesterday's economic price is actually observable under the same clock. The limitation is explicit, and all models share the same information set.

HistoricalMeanDelta adds the fitted mean residual at each horizon to a[t]. This controls for a simple average correction to the anchor. Ridge uses all 32 standardized current-row predictors and a linear multi-output regression of the same standardized residuals, with an unpenalized intercept. Alpha is chosen among 1, 10, 100, 1,000 and 10,000 using validation standardized MSE; the chosen alpha is 1,000. It is refit on the same pre-test label paths. Ridge can represent linear information in current levels; it lacks the LSTM's full sequence and nonlinear gates.

The published-WTI-only LSTM uses the selected architecture and fixed duration with just the anchor sequence and seed 42. It tests whether complex predictors seem useful descriptively. It is not separately optimized, and its single-seed comparison with a three-seed main ensemble is not a controlled feature-importance experiment. Teammates' ARIMA/VAR/tree models are not reimplemented here; compare them by common exported origins/targets/horizons.

9. Evaluation equations and interpretation

MAE = mean(abs(yhat-y)); RMSE = sqrt(mean((yhat-y)^2)). Both use USD/barrel; RMSE penalizes large misses more strongly. Percentage improvement is 100*(1 - model_error/persistence_error). Negative values mean worse performance.

The supplementary MAPE = 100*mean(abs(yhat-y)/y) is restricted to y>1, with `mape_n` reported. This avoids undefined or misleading divisions around zero and negative prices but changes the eligible sample. Do not compare MAPE across models using different filtered rows.

Directional accuracy compares sign(yhat-a) and sign(y-a), excluding actual unchanged prices. Up precision is TP/(TP+FP), up recall is TP/(TP+FN), and F1 is their harmonic mean. When a class denominator is zero, the code reports zero. The neutral forecast share is also exported. Persistence's neutral forecasts are counted as wrong on nonzero actual moves; it should be described as abstaining, not as a 0%-skill direction classifier. This published-anchor measure is clearly distinguished from close-to-close direction.

For each horizon, all models share the same observed date pairs. The test contains 689 horizon-1 origins and 630 horizon-60 origins. Besides horizon-specific full samples, a common-origin table uses 630 complete paths. Predictions on neighboring days and horizons overlap; 630 forecasts are not 630 independent 60-day experiments.

Robustness is summarized by origin year and origin OVX above/below the fitted pre-test median. The bootstrap resamples paired model/benchmark errors in 60-origin blocks and recomputes RMSE improvement. Its ranges are approximate descriptive uncertainty under a dependence-preserving resampling choice. All four selected-horizon intervals include zero; no statistically secure superiority is claimed. No test-based architecture edits were made to improve these reported results.

10. Code map and how the pieces connect

`read_data` checks chronological dates and obtains permitted predictors from the dictionary. `boundary` imposes a release buffer. `origins_between` enforces whole-path label cutoffs. `windows` builds causal windows ending at the origin. `targets` builds the 60 dollar deltas. `fit_scaler` estimates only on the supplied fitting sample.

`OilLSTM` defines the network. `train_one` fits it with gradient clipping and optional validation early stopping. `predict` performs batch inference without gradients. `ridge_fit` and `ridge_predict` implement the regularized linear benchmark without an additional machine-learning dependency. `metric_values`, `evaluate` and `bootstrap_ci` generate the comparisons. `make_plots` exports the figures.

`run` orchestrates validation, final refitting, frozen testing and output generation. `forecast_from_checkpoint` loads saved models/scalers and needs only predictor rows. `main` exposes the command-line interface. Test cases cover feature exclusion, split purging, window causality, horizon alignment, negative-price retention, constant scaling and invariance of forecasts when every target is removed.

11. Automatic updates and operational boundaries

A new processed row triggers inference through `--forecast-only`. Training is not required for every update. The saved feature list/order, scalers, architecture and seeds are checked/loaded; the final lookback window yields 60 outputs. The current target can be blank or absent. A meaningful update needs genuinely newer features; rerunning unchanged data reproduces the same forecast. There is no persistent network request, recurring scheduler or upstream GitHub integration in this package because no repository/API endpoint was supplied.

The saved model was fitted with labels through 14 December 2023 and is intentionally frozen throughout the supplied historical test. A deployment refit can be added on a chosen cadence after reconstructing a new train/validation cutoff using actually released labels. Such a refit must be evaluated as its own walk-forward experiment. Do not overwrite the original test evidence and then describe it as untouched.

12. Limits and defensible next research

The data vintage and upstream alignment are not independently verifiable. Ten observations of boundary buffering are conservative but cannot certify all historical publication delays. The test includes large changes in 2026; this frozen model's gains are concentrated there and do not generalize uniformly to earlier years. The three-seed ensemble measures only initialization variation, not all model uncertainty. No calibrated future-price intervals are provided.

WTI spot quotes are not an executable portfolio. The supplied file has no matched instrument opening/execution prices, futures contract roll history, margin assumptions or cost schedule. Therefore this contribution does not fabricate a Sharpe ratio or P&L from forecast errors. If the group adds economic-value analysis, map time-stamped signals to an actual traded instrument using an agreed execution delay, returns, rolls and costs, then evaluate on a genuinely held-out period. A simple sign(yhat-a) signal alone is not a valid backtest.

Future work can test expanding/rolling refits, a larger but validation-only search, recent target lags when release-time evidence permits them, and ensembles with weights selected before test. An additional untouched period is necessary if these changes are motivated by the current results. Consistent improvements across years, errors and meaningful benchmarks would be stronger evidence than an isolated low RMSE.

References

1. Supplied QF5210 Part II introduction PDF, oil forecasting brief on PDF pages 8-9 and presentation rules on page 10.

2. Supplied `data/features.csv`: exact variable definitions, units and availability assumptions.

3. PyTorch LSTM documentation: https://docs.pytorch.org/docs/stable/generated/torch.nn.modules.rnn.LSTM.html

4. U.S. EIA, WTI Cushing spot series via FRED DCOILWTICO: https://fred.stlouisfed.org/series/DCOILWTICO

5. Alquist, Kilian and Vigfusson (2011), Forecasting the Price of Oil, Federal Reserve IFDP 1022: https://www.federalreserve.gov/pubs/ifdp/2011/1022/ifdp1022.htm . This provides context for oil forecast benchmarks; its empirical results are not attributed to this project.

What to present and how to explain it

Recommended speaking allocation

The group has eight minutes in total plus two minutes of Q&A. Your individual LSTM segment should usually take 90-120 seconds; agree the exact allocation with teammates. Use deck slides 1-3 as the main segment and slides 4-6 only as backups. Do not present this entire technical report during the short group talk.

Main slide 1: method and fairness (about 35 seconds)

Say: "My contribution is a multivariate LSTM for WTI forecasting. It takes the previous 60 rows of already published energy and financial information and directly predicts all 60 future horizons. We forecast dollar changes from the last published WTI quote, then add that quote back to obtain prices. We exclude the target price from the inputs, use chronological splits, and select the architecture and training duration on validation data only."

Point to the input/output dimensions and split dates. Explain that the available weekly quote may be stale. Avoid reciting every predictor or every gate equation in the main talk. If asked why h=1 is the origin date, explain the before-midnight forecast clock.

Main slide 2: actual comparative result (about 35 seconds)

Say: "On the 2024 to October 2026 test sample, the LSTM gives almost no improvement at the shortest horizon. At horizon 60 its RMSE falls from 15.56 dollars per barrel for published-price persistence to 14.29, an 8.1 percent improvement. However, its MAE stays near 10.34, and ridge regression has lower RMSE and MAE. So the LSTM captures some information that reduces large misses, but it is not our strongest model overall."

Use the key-horizon table or the editable RMSE chart. Lower is better. Do not show only the actual-versus-predicted curve: persistent oil prices can make a weak forecast look visually convincing. Do not say an 8.1% RMSE gain means 8.1% trading return, 91.9% accuracy, or a probability of correctness.

Main slide 3: robustness and operational output (about 30 seconds)

Say: "The gain is concentrated in high-volatility conditions and 2026. At horizon 60 the LSTM is worse than persistence for 2024 and 2025 origins. A block bootstrap gives a broad improvement range including zero, so we do not claim statistically secure superiority. The code saves date-aligned forecasts and can update them immediately when a new processed feature row is appended. These outputs can be compared with the other models or used in a validation-weighted ensemble."

Finish by stating your contribution specifically: "I implemented the LSTM, the causal multi-horizon data pipeline, validation selection, benchmark evaluation, robustness checks and forecast export." Use this only after reviewing and understanding the supplied implementation; follow course disclosure rules for assistance.

What each output is for

Output	Use in the presentation or group submission
`outputs/accuracy_by_horizon.png`	Full error curves; shows whether gains persist across horizons
`outputs/test_metrics_key_horizons.csv`	Exact values for a concise comparison table
`outputs/forecast_vs_actual.png`	Illustrates large misses and response to different periods
`outputs/robustness.csv`	Year and origin-OVX comparisons for the robustness slide
`outputs/bootstrap_improvement.csv`	Explains why a point improvement is uncertain
`outputs/latest_forecast_path.png`	Optional demonstration of the 60-output system; label historical origin explicitly
`outputs/test_predictions.csv`	Machine-readable join with teammates' model forecasts
`outputs/models/`	Reuse the model without training; not a presentation graphic
`Oil_LSTM_Walkthrough.ipynb`	Rehearsal, code explanation and reproducible demonstration

Questions your professor may ask

Why LSTM rather than ARIMA or a linear model?

LSTM can learn nonlinear relationships and temporal dependencies in multiple predictors. That is why it is worth testing, not why it must win. ARIMA and linear models are often competitive on small, noisy financial datasets. Our own ridge comparison reinforces that point.

What is the memory mechanism?

Cell memory is updated by forget and input gates; the output gate controls the hidden representation. The last hidden state summarizes the observed window, and a linear layer maps it to 60 standardized forecast deltas. Have the gate equations in the technical report ready as backup.

How exactly do you avoid leakage?

The target is not an input. Windows end at the origin. The features follow the supplied publication-aligned dictionary. Scalers use only the fitting sample. All 60 labels of a training or validation sample must lie before its permitted label cutoff. Ten additional observation rows are withheld before boundaries. Test labels never select hyperparameters or duration. Upstream vintage/release correctness remains an explicit assumption because raw sources and preprocessing were not supplied.

Is a current-row input already a leak?

Not under the supplied dictionary: current-row predictors were known before that day's 00:00 New York forecast cutoff. The current-row target was not known and is excluded. If a teammate's pipeline instead treats rows as after-close information, the alignment must be changed consistently before comparison.

Why not use yesterday's WTI target as a lag?

Economic date and publication date differ. The dictionary says the target is unavailable at the forecast cutoff; it supplies an explicitly released WTI quote instead. A lag of the target would be allowed only after verifying the release timing for that lag.

Does 60 mean calendar days?

It means 60 observed trading rows, including the upcoming origin-date observation. We do not create holiday prices. State this convention upfront and ensure the other models use it. If the group interprets 1-60 as calendar days, the targets/calendar must be rebuilt for every model rather than relabeled.

Why use dollar residuals and not log returns?

The data includes a negative WTI price in 2020, so log prices and log returns are not generally defined. Dollar residuals from an available quote avoid that problem and retain the extreme event.

Why is MAPE not the headline metric?

MAPE becomes undefined or unstable around zero and is difficult to interpret for negative prices. We lead with MAE and RMSE. The supplementary MAPE uses only actual prices above USD 1 and reports its eligible count.

Why only four epochs?

Four was the best validation checkpoint for the selected candidate. More epochs worsened validation loss. The final refit uses the selected duration with fresh initializations. We did not choose the epoch using the test error. Four is specific to this setup and should not be described as a universal optimum.

Why average three seeds?

It reduces dependence on one random initialization. Seeds and equal weights were fixed in advance. Their spread does not include every source of forecast uncertainty and is not a confidence or prediction interval.

Why is short-horizon error quite large despite many predictors?

The reference WTI quote is delayed, and oil can move sharply before the forecast day's target. Inputs also include slow-moving weekly values. The model therefore solves a harder available-information problem than one that assumes yesterday's final price is already known. The October 2026 example illustrates this limitation.

Is the 8.1% improvement statistically significant?

We do not claim that. The approximate block-bootstrap range for horizon-60 improvement is -11.72% to +13.89%, including zero. Overlapping horizons reduce the effective information in the sample. Regime dependence and limited blocks further weaken broad conclusions.

Why can RMSE improve while MAE does not?

RMSE gives large errors extra weight, so a model can reduce a few large misses while making typical smaller forecasts slightly worse. MAE reflects average absolute error. Both must be reported to avoid a one-sided conclusion.

What does directional accuracy mean here?

It is the sign of the target's change from the latest published quote known at the origin. It is not the next close-to-close return. Persistence issues a neutral view and should not be treated as a meaningful 0%-accurate direction classifier. Our main LSTM direction scores around 48-51% at the chosen horizons do not establish strong directional skill.

Can we trade profitably from this?

The current experiment does not establish that. The dataset contains spot quotes, not matched executable trade prices, rolls or costs. A proper futures/ETF backtest would require a separate execution design. Forecast-error improvement alone is not a profitable strategy.

How does it update tomorrow?

Append a new correctly aligned feature row, including the known released WTI anchor, and run the forecast-only command. It loads saved weights/scalers and emits 60 new forecasts. The new target can be unknown. Data downloading and scheduled execution would be added around this process after the upstream source is identified.

How should the group combine models?

First join the exact origin-date/target-date/horizon rows and compare on their intersection. If ensembling, choose weights on validation predictions with the same timing convention. Do not choose weights from this test table or fill missing forecasts with another model's target values.

A short message you can copy to your teammate

"My LSTM component is complete in the attached folder. It forecasts WTI at horizons 1-60 from publication-aligned predictors, with chronological validation and a frozen 2024-onward test. At h=60 the LSTM RMSE is 14.29 versus persistence 15.56, but ridge is better at 13.96 and the LSTM improvement is not robust across years. Please use the exported origin/target/horizon keys for comparisons and keep the forecast clock consistent. The package includes all code/data, saved models, results, a technical report, editable slides and speaking notes. I suggest about 90-120 seconds for my three main slides. Please retain the uncertainty and provenance limitations in the final group discussion."

Teammate integration and submission

Contract for model comparisons

The LSTM requires only the processed predictors, not another model's results. The test is frozen, with origins beginning 2024-01-02 and final observed target date 2026-10-06. Model selection uses data before 2024, with the stricter buffer and full-path purging described in `methodology.md`.

The prediction key is `(origin_date, target_date, horizon, model)`. At an origin before the row's trading day, h=1 targets that row; h=k targets row t+k-1. Horizons count observed rows. All raw results are in USD/barrel. The anchor is the last released WTI quote, not necessarily the previous observation's target. Both dictionary and data are included unchanged.

To compare with another model, select a common horizon and inner-join the exact keys except model. Check the actual target values agree. Compare on the shared rows and report counts. Do not merge by target date alone: multiple origins can predict a given target date. Do not align h=1 from an after-close model with h=1 from this before-midnight model without shifting the origin convention correctly.

```
import pandas as pd
lstm = pd.read_csv('outputs/test_predictions.csv')
lstm = lstm[lstm.model.eq('LSTM')]
other = pd.read_csv('teammate_predictions.csv')
keys = ['origin_date', 'target_date', 'horizon']
paired = lstm.merge(other, on=keys, how='inner', suffixes=('_lstm', '_other'), validate='one_to_one')
assert (paired.actual_lstm - paired.actual_other).abs().max() < 1e-8
```

This assumes the other file contains one chosen model. Filter it first if it contains multiple models. The group's original train/test arrangement was not supplied; the dates used here are a documented default. If the group requires different dates, rerun all models consistently and regenerate the figures, report and slide values. Do not paste mismatched test results into one comparison table.

Files to include in the final group ZIP

File/folder	Purpose
`data/wti_daily.csv`, `data/features.csv`	Required daily observations and full variable definitions
`src/oil_lstm.py`	Full training, selection, evaluation and forecast code
`requirements.txt`, `environment_tested.txt`	Install instructions and tested versions
`tests/`, `outputs/verification.log`	Checks for timing and label leakage
`Oil_LSTM_Walkthrough.ipynb`	Executed explanatory notebook
`outputs/test_predictions.csv`	Row-level model predictions for integration
`outputs/test_metrics_*.csv`	Tables at every horizon and common origins
`outputs/robustness.csv`, `outputs/bootstrap_improvement.csv`	Stability and uncertainty evidence
`outputs/models/`	Three checkpoints, fitted scalers and model metadata
`outputs/*.png`	Figures for report and discussion
`LSTM_presentation.pptx`	Editable main/backup slides with speaker notes
`output/pdf/LSTM_technical_report.pdf`	Detailed methodology and results
`presentation_script_and_QA.md`	Talk script and professor questions
`build_handoff.py`	Regenerates data-dependent documentation/notebook/PDF

`environment_tested.txt` records relevant package versions for this run. The full machine environment is not bundled. Presentation source/export is supplied for editing; model reproduction does not depend on the presentation authoring tool.

Submission and contribution

The supplied brief specifies a ZIP including editable presentation slides, data and code, with optional prerecorded video, due 31 October 2026. Confirm any amendments in Canvas. The LSTM presentation is one member's section, not a replacement for the group's eight-minute combined deck. Replace the contribution line with the student's name in the group's final material, retain references and use any required AI-assistance disclosure.

Suggested contribution text after you have reviewed the work: "LSTM specification, multi-horizon causal window construction, chronological validation, forecast benchmarking, robustness analysis, reproducible code and results export." Do not claim that the raw-data processing or teammates' methods were your work.

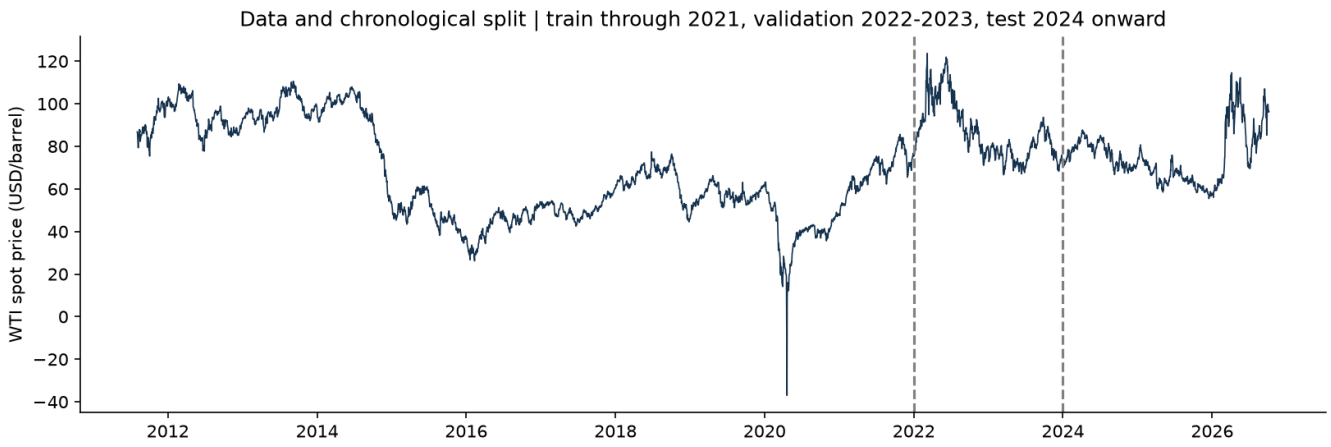
When results change

Keep the supplied output folder as the original experiment. New split dates, input revisions, altered features or hyperparameters require a new result folder and matching slides. `build_handoff.py` currently reads `outputs/`; update its source path or replace that folder only after preserving the old run. Its narrative is written for the supplied baseline experiment, so review all fixed dates, numbers and interpretations when rebuilding a different experiment. Automatically regenerated tables alone do not validate old prose.

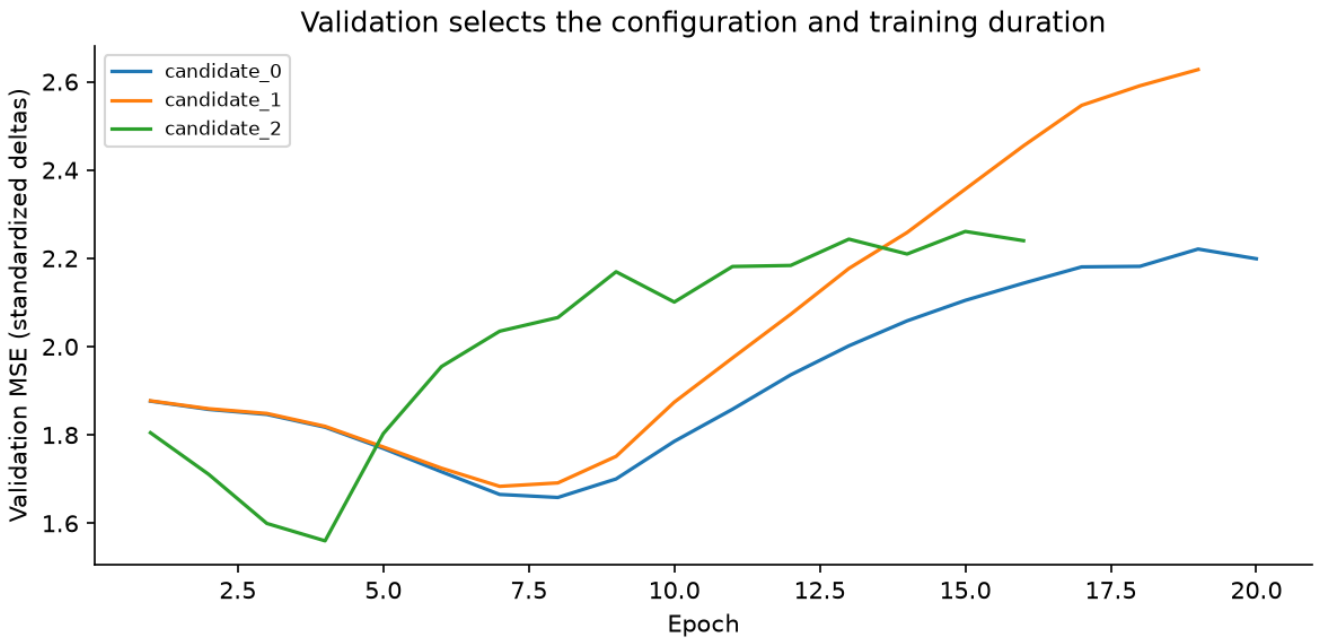
If you want a live forecasting system, connect the teammate's original GitHub preprocessing to the forecast-only CLI and trigger it after each validated data refresh. Historical target vintages and release times need an upstream audit. Do not silently use newer revised data and describe the result as a fully real-time test.

Figure appendix

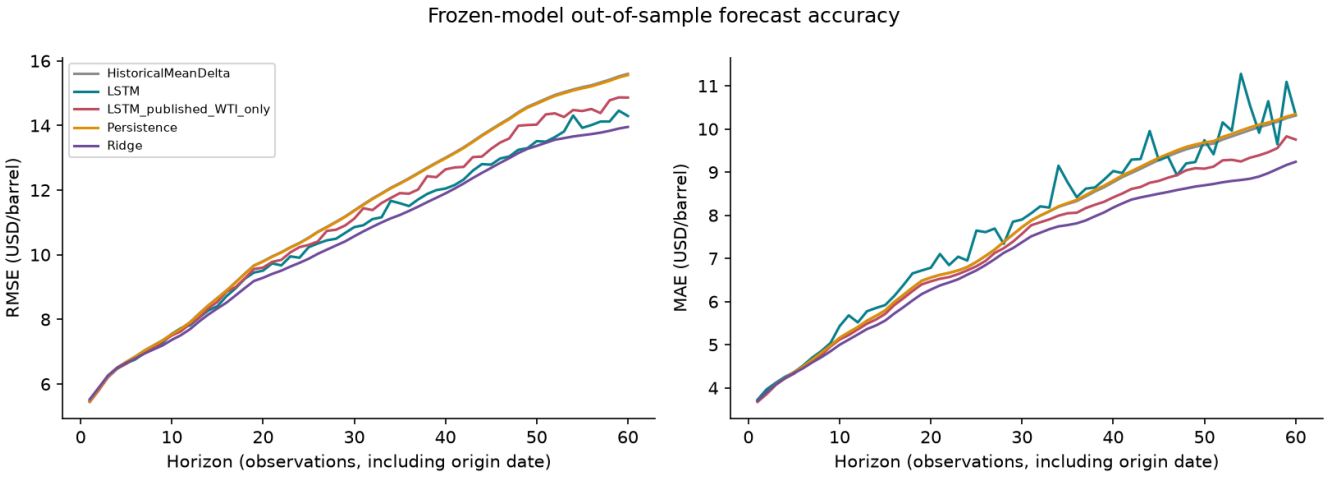
Chronological data and split boundaries



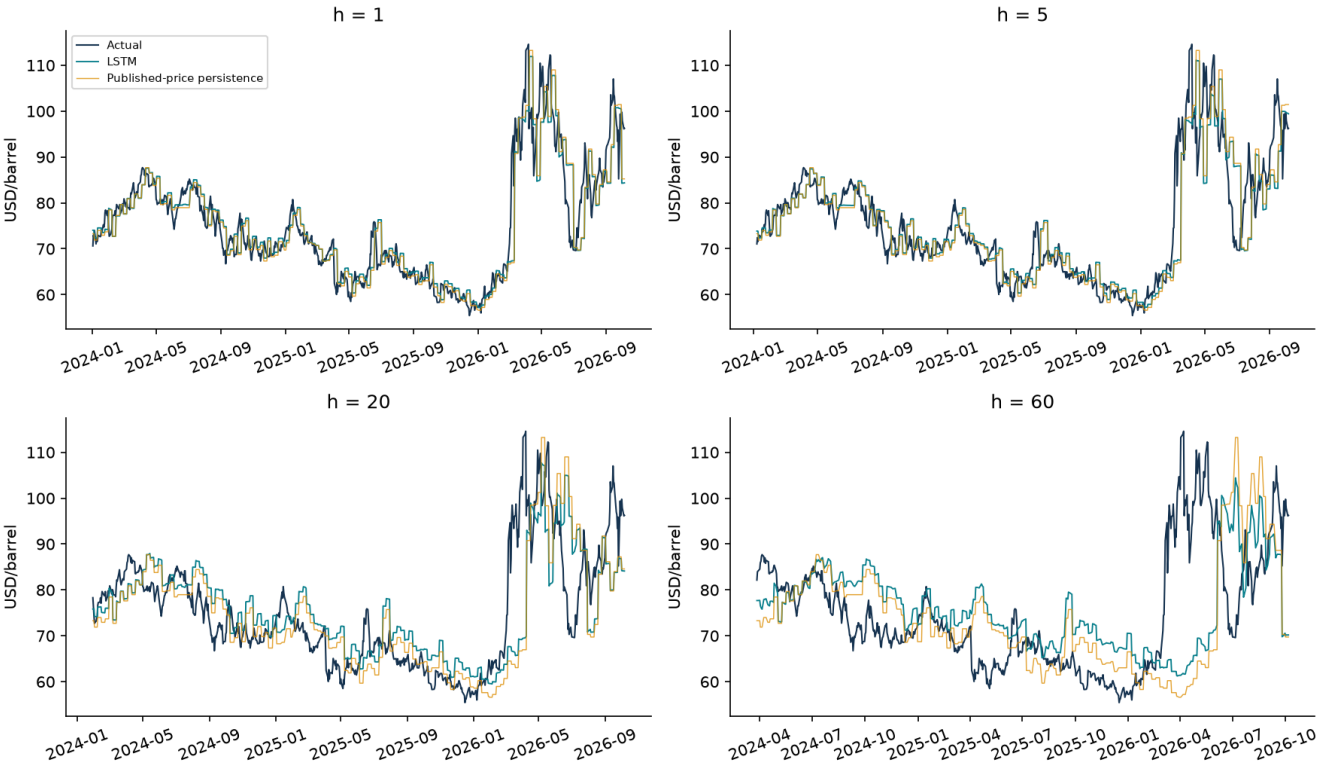
Validation learning curves for the three candidates



Test MAE and RMSE over all 60 horizons



Date-aligned test forecasts at selected horizons



Historical last-origin path; seed range is not a prediction interval

